Final project Self-assessment report

Team: <23120111>-<23120124>-<23120175>

GitHub repo URL: <https://github.com/JakeConal/smart-restaurant-admin>

TEAM INFORMATION

Student ID	Full name	Git account	Contribution	Contribution percentage (100% total)	Expected total points	Final total points
23120124	Nguyễn Minh Hiếu	JakeConal	Team Lead & Customer App Architect. Engineered the end-to-end Dine-in Ordering flow, WebSocket Tracking, Fuzzy Search, and high-performance Client-side Caching.	35%	10	
23120175	Huỳnh Thái Toàn	Toanapr	System Architect & Payment Specialist. Designed the Global Database Schema, RBAC system, VNPay Integration, and Kitchen Display System (KDS). Implemented secure JWT Auth via HttpOnly cookies.	34%	10	
23120111	Phạm Quốc Nam Anh	kyrie25	Backend Specialist & Reporting Lead. Developed the Sales & Revenue reporting engine with interactive visualizations. Modularized backend architecture and implemented a localized QR generation system.	31%	10	

FEATURE LIST

Project: Smart Restaurant - QR Menu Ordering System

Students must input minus points to every uncompleted feature in the SE column.

*SE: Self-evaluation

*TR: Teacher review

ID	Features	Grade			Notes
		Point	**SE***	**TR***	
1	**Overall requirements**				
	User-centered design	-5	0		Built with user experience in mind, not just feature list. Focus on solving real restaurant problems: seamless QR ordering, efficient waiter workflow, real-time kitchen coordination, and convenient payment options
	Database design	-1	0		Database with tables: users, restaurants, menus, menu_items, categories, modifiers, tables, orders, order_items, payments
	Database mock data	-1	0		Sample restaurants, menu items, categories, tables, and test orders
	Website layout	-2	0		Two layouts: Customer mobile ordering interface and Admin dashboard
	Website architect	-3	0		Based on MVC architecture. Clear separation of concerns with controllers, services, repositories. Client-side validation, Input validation, Business rule validation
	Website stability and compatibility	-2	0		Mobile-first responsive design, tested on Chrome and Safari
	Document	-1	0		Clear documentation for developers and users: setup guide, API endpoints, database design, system architecture, user guide
	Demo video	-5	0		Video demonstrating all features: restaurant signup, menu management, QR ordering, payment, KDS
	Publish to public hosts	-1	0		Deployed to a public hosting service with accessible URL
	Development progress is recorded in Github	-7	0		Git history with meaningful commits, branches for features, pull requests
2	**Guest features (Customer Ordering)**				

ID	Features	Grade		Notes
	Home page (Menu page)	-0.25	0	Restaurant menu page loaded via QR code scan with categories and items
	View list of menu items	-0.25	0	Display menu items with images, prices, descriptions
	Filter menu items by			A combination of the criteria
	› Item name	-0.25	0	Search menu items by name
	› Category	-0.25	0	Filter by food categories (Appetizers, Main Dishes, Drinks, Desserts)
	Sort menu items by popularity	-0.25	0	Sort by most ordered items
	› Chef recommendation	-0.25	0	Filter/highlight items marked as chef's recommendations
	Menu item paging	-0.75	0	Pagination for large menus with infinite scroll. URL updated on search/filter/paging
	View menu item details	-0.25	0	Item detail page with full description, modifiers, allergen info
	View menu item status	-0.25	0	Display item availability status (Available, Unavailable, Sold out)
	Show related menu items	-0.25	0	Suggest items from same category or popular pairings
	View list of item reviews	-0.5	0	Customer reviews for menu items with pagination
	Add a new item review	-0.25	0	Logged-in customers can review items they ordered
	Shopping cart (Order Cart)			
	› Add a menu item to the Cart	-0.25	0	Add items with quantity selection
	› View and update items in the Cart	-0.5	0	Cart summary with items, quantities, modifiers, prices. Update quantity with auto-update totals
	Ordering and payment (Dine-in)			

ID	Features	Grade		Notes
	› Bind the shopping cart to the table session	-0.25	0	Cart persists for table session
	› Input order details (notes, special requests)	-0.25	0	Guest name, special instructions field
	› Add items to current order	-0.25	0	Customers can add more items to their unpaid order (single order per table session)
	› View order status	-0.25	0	Guest can track order status (Received → Preparing → Ready)
	› View order details	-0.25	0	Order confirmation with items, total, table number
	› Request bill	-0.25	0	Customer requests bill when ready to pay
	› Process payment after meal	-0.25	0	Stripe payment processing after dining
3	**Authentication and authorization**			
	Use a popular authentication library	-1	0	Passport.js with JWT strategy
	Registration (Customer Signup)	-0.5	0	Customer registration with email/password. Real-time email availability check
	Verify user input: password complexity, full name	-0.25	0	Password rules, required fields validation
	Account activation by email	-0.25	0	Email verification link sent on signup
	Social Sign-up/Sign-In	-0.25	0	Google OAuth integration
	Login to the website	-0.25	0	JWT-based authentication for admin/staff

ID	Features	Grade		Notes
	Authorize website features	-0.25	0	Role-based access control (Admin, Waiter, Kitchen Staff, Customer)
	Forgot password by email	-0.25	0	Password reset via email link
4	**Features for logged-in users (Customers)**			
	Update user profile	-0.25	0	Customer can update name, preferences
	Verify user input	-0.25	0	Input validation on profile updates
	Update the user's avatar	-0.25	0	Profile photo upload
	Update password	-0.25	0	Change password with old password verification
	Order history and tracking			
	› View order history	-0.25	0	List of past orders linked to user account
	› View item processing status	-0.25	0	Track individual item status within an order (Queued, Cooking, Ready)
	› Real-time Order Updates	0.5	+0.5	WebSocket-based live order status updates for customers
5	**Administration features (Restaurant Admin)**			
	Create Admin accounts	-0.25	0	Admin creates additional Admin accounts
	Manage Admin accounts	-0.25	0	View, edit, deactivate Admin accounts
	Update admin profile	-0.25	0	Restaurant admin profile management
	Create Waiter accounts	-0.25	0	Admin creates accounts for waiters

ID	Features	Grade		Notes
	Create Kitchen Staff accounts	-0.25	0	Admin creates accounts for kitchen staff
	Manage menu categories	-0.25	0	Create, edit, delete food categories
	View menu item list	-0.5	0	List all menu items with filters and pagination
	Filter menu items by name, category	-0.25	0	Search and filter menu items
	Sort menu items by creation time, price, popularity	-0.25	0	Sortable menu item list
	Create a new menu item	-0.25	0	Add item with name, price, description, category, prep time
	Upload multiple menu item photos	-0.5	0	Multi-image upload for menu items
	Add menu item to category with modifiers	-0.25	0	Assign categories and create modifier groups (Size, Extras)
	Menu Item Modifiers	0.5	+0.5	Modifier groups (Size, Extras) with price adjustments
	Specify menu item status	-0.25	0	Available, Unavailable, Sold out
	Verify user input	-0.25	0	Input validation for menu items
	Update a menu item	-0.25	0	Edit existing menu items
	Add, remove menu item photos	-0.25	0	Manage item images
	Change menu item category, modifiers	-0.25	0	Update item categorization
	Update menu item status	-0.25	0	Toggle availability
	Verify user input	-0.25	0	Validation on updates

ID	Features	Grade		Notes
	Customer orders (Order Management)			
	› View list of orders sorted by creation time	-0.25	0	Order list in KDS sorted by time
	› Filter orders by status	-0.25	0	Filter: Received, Preparing, Ready, Completed
	› View order details	-0.25	0	Full order details with items, modifiers, notes
	› Update order status	-0.25	0	Progress order through states: Received → Preparing → Ready → Completed
	› Kitchen Display System (KDS)	-0.5	0	Real-time order display for kitchen staff with sound notifications
	› Order Timer and Alerts	-0.25	0	Highlight orders exceeding item's configured prep time
	Table Management			
	› Create, edit, deactivate tables	-0.5	0	Create, edit, deactivate tables with capacity and location
	› QR Code Generation	-0.5	0	Generate unique QR codes per table with signed tokens
	› QR Code Download/Print	-0.25	0	Download QR as PNG/PDF for printing
	› QR Code Regeneration	-0.25	0	Regenerate QR and invalidate old codes
	Reports			
	› View revenue report in time range	-0.25	0	Daily, weekly, monthly revenue reports
	› View top revenue by menu item in time range	-0.25	0	Best-selling items report
	› Show interactive chart in reports	-0.25	0	Chart.js/Recharts for analytics dashboard (orders/day, peak hours, popular items)

ID	Features	Grade		Notes
7	**Waiter features**			
	View pending orders	-0.25	0	List of new orders waiting for waiter acceptance
	Accept/Reject order items	-0.25	0	Waiter can accept or reject individual order items
	Send orders to kitchen	-0.25	0	Forward accepted orders to Kitchen Display System
	View assigned tables	-0.25	0	See tables assigned to the waiter
	Mark orders as served	-0.25	0	Update order status when food is delivered to table
	Bill Management			
	› Create bill for table	-0.25	0	Generate bill with all order items, subtotal, tax, and total
	› Print bill	-0.25	0	Print bill to thermal printer or download as PDF
	› Apply discounts	-0.25	0	Apply percentage or fixed amount discounts to bill
	› Process payment	-0.25	0	Mark bill as paid (cash, card, or e-wallet)
8	**Advanced features**			
	Payment system integration	0.5	+0.5	Payment gateway integration (ZaloPay, MoMo, VNPay, Stripe, etc.) - at least 1 required
	Fuzzy search	0.25	+0.25	Fuzzy matching for menu item search with typo tolerance
	Use memory cache to boost performance	0.25	+0.25	Redis for menu caching and session management
	Analyze and track user actions	0.25	0	Google Analytics for QR scan tracking, order conversion metrics
	Dockerize your project	0.25	0	Docker containers for backend, frontend, database
	CI/CD	0.25	0	GitHub Actions for automated testing and deployment

ID	Features	Grade		Notes
	Monitoring and logging	0.25	0	Centralized application logs, metrics, dashboards, and alerting (e.g., ELK/EFK, Prometheus/Grafana)
	BI integration	0.25	0	Connect operational data to BI tools for reporting and dashboards (e.g., Power BI, Tableau, Metabase)
	Advanced authorization (RBAC)	0.25	+0.25	Fine-grained role/permission management for Admin/Chef/Waiter and other staff roles
	WebSocket real-time updates	0.5	+0.5	Socket.IO for real-time features: KDS order notifications, customer order status tracking, waiter new order alerts, kitchen ready notifications, table status updates
	Multi-tenant support	0.5	0	Multiple restaurants (tenants) with strict data isolation; tenant-scoped RBAC and configuration
	Multilingual support	0.25	0	i18n for English/Vietnamese language selection

GIT HISTORY

Contributors

Avatar	Username	Commits	Additions	Deletions
	JakeConal	87	142,500+	78,200+
	Toan Huynh	79	35,800+	15,400+
	Nam Anh	12	19,500+	45,200+

Commits

List significant commits here with format:

Date	Author	Commit Message	Files Changed
2026-01-16	Toan Huynh	feat: Implement VNPay IPN endpoint and refactor payment integration to use numeric order IDs.	2
2026-01-16	Toan Huynh	feat: Set default payment method to 'vnpay' and broadcast order updates after marking orders as paid.	1

Date	Author	Commit Message	Files Changed
2026-01-15	JakeConal	feat: implement menu caching mechanism to enhance performance and reduce API calls.	4
2026-01-15	Toan Huynh	feat: Implement VNPay payment gateway with dedicated frontend return page and backend services for payment creation and verification.	8
2026-01-15	Nam Anh	feat: add comprehensive sales revenue reporting page.	11
2026-01-15	Toan Huynh	feat: Implement Super Admin functionality with admin management features.	23
2026-01-15	Toan Huynh	feat: Implement refresh token functionality with secure cookie management.	7
2026-01-14	JakeConal	feat: implement restaurant-specific WebSocket connections and real-time order handling.	16
2026-01-14	JakeConal	feat: add Bill function for invoice preview and printing.	17
2026-01-14	JakeConal	feat: add kitchen sound notification and highlight orders exceeding prep time.	8
2026-01-14	Toan Huynh	feat: add kitchen staff and waiter management functionality.	25
2026-01-13	JakeConal	feat: add functionality to send and receive orders via websocket.	38
2026-01-12	Toan Huynh	feat: Implement Waiter Tables feature with table assignment and occupancy status.	8
2026-01-11	Toan Huynh	feat: implement email verification flow and password reset functionality.	10
2026-01-10	Toan Huynh	feat: implement admin authentication module with role-based access control and permissions.	45
2026-01-10	JakeConal	feat: apply fuzzy search in customer app for menu item Discovery.	4
2026-01-09	JakeConal	feat: Implement order tracking and payment functionality for customer app.	32
2026-01-08	JakeConal	feat: implement review module with CRUD operations and integrate with frontend.	15
2026-01-08	JakeConal	feat: enhance cart and menu pages with improved UI, URL state management, and item details display.	20

Date	Author	Commit Message	Files Changed
2026-01-07	JakeConal	feat: Revamp Menu and Bottom Navigation UI with modern design elements.	15
2025-12-31	JakeConal	feat: implement photo upload and retrieval for menu items and integrate Google Login in Admin app.	25
2025-12-27	Nam Anh	perf(table): use local QR generator and modularize backend API architecture.	27
2025-12-25	JakeConal	feat: implement MenuCategory, MenuItem, and Authentication modules with JWT.	39
2025-12-19	JakeConal	feat: initial backend setup with NestJS and TypeORM integration for table management.	42
2025-12-18	JakeConal	feat: initialize smart restaurant project with Next.js, React, and Tailwind CSS.	17
2026-01-14	JakeConal	feat: add kitchen sound notification and highlight orders exceeding item's prep time	8
2026-01-14	JakeConal	feat: implement restaurant-specific WebSocket connections and order handling	16
2026-01-14	JakeConal	feat: update role-based access for WaiterOrdersPage and Sidebar to include ADMIN role	2
2026-01-14	Nam Anh	feat(backend): add example env	1
2026-01-14	Nam Anh	chore: remove conflicting lock file	1
2026-01-14	Toan Huynh	fix(eslint): update ESLint configuration to include TypeScript parser and options	1
2026-01-14	Toan Huynh	refactor: remove avatar_url from KitchenStaff and Waiter interfaces and related components	8
2026-01-14	Toan Huynh	feat: add kitchen staff management feature	12
2026-01-14	Toan Huynh	feat: add waiter management functionality	13
2026-01-14	Toan Huynh	fix(sidebar): restrict waiter orders and tables visibility to WAITER role only	1
2026-01-13	JakeConal	feat(order): add special requests field and update related components for better order handling	5

Date	Author	Commit Message	Files Changed
2026-01-13	JakeConal	feat(payment): enhance payment flow with support for multiple unpaid orders	3
2026-01-13	JakeConal	feat(kitchen): implement kitchen order management and status updates	18
2026-01-13	JakeConal	refactor: remove unused components and modals	39
2026-01-13	JakeConal	feat: add functionality to send and receive orders via websocket.	38
2026-01-12	JakeConal	feat: Implement soft delete for menu categories and add remove functionality	3
2026-01-12	JakeConal	feat: Add CustomerJwtStrategy for handling customer JWT authentication	3
2026-01-12	JakeConal	feat: Enhance popularity score management by adding update functionality and adjusting scores on review changes	3
2026-01-12	JakeConal	feat: create Tailwind CSS configuration with custom theme and animations	13
2026-01-12	Toan Huynh	feat: Implement role-based redirection in login and enhance waiter pages with improved layout	4
2026-01-12	Toan Huynh	feat: Update Sidebar to display user role dynamically and restrict admin menu items	1
2026-01-12	Toan Huynh	feat: Implement Waiter Tables feature with table assignment and occupancy status	8
2026-01-12	Toan Huynh	chore: Remove outdated Waiter Features & Bill Management implementation roadmap	1
2026-01-12	Toan Huynh	feat(backend): Add UsersController for managing waiters with permission checks	1
2026-01-12	Toan Huynh	docs: Add Phase 1 waiter features roadmap and update project files	5
2026-01-12	Toan Huynh	fix(frontend): Add API proxy, update navigation, and fix decimal field type handling	5
2026-01-12	Toan Huynh	feat(backend): Add database seeding for tables and orders with user-specific data	9
2026-01-12	Toan Huynh	feat(phase1): Add polling hooks for real-time order updates and offline queue processing	4

Date	Author	Commit Message	Files Changed
2026-01-12	Toan Huynh	feat(phase1): Implement waiter dashboard UI components and manager escalation interface with bento design	11
2026-01-12	Toan Huynh	feat(phase1): Implement waiter and manager API endpoints with order acceptance and escalation	11
2026-01-12	Toan Huynh	feat(phase1): Add order and table schemas with waiter assignment and escalation fields	4
2026-01-11	Toan Huynh	fix: update apostrophe syntax and improve email verification link formatting	2
2026-01-11	Toan Huynh	fix: update apostrophe syntax in email verification messages	1
2026-01-11	Toan Huynh	feat(auth): implement email verification flow with resend option and password reset functionality	10
2026-01-10	Toan Huynh	feat: Refactor schema imports and add role-based access control entities	11
2026-01-10	Toan Huynh	feat: Implement email verification and password reset functionality for admin users	13
2026-01-10	JakeConal	feat: apply fuzzy search in customer app	4
2026-01-10	Toan Huynh	feat: Refactor JWT authentication to use ConfigService for secret management and streamline guard implementations	5
2026-01-10	Toan Huynh	feat: Enhance admin authentication with role-based access control and permissions	15
2026-01-10	Toan Huynh	feat: Add database schema for user roles, permissions, and credentials	6
2026-01-10	Toan Huynh	feat: Implement admin authentication module with login and signup functionality	30
2026-01-09	JakeConal	feat: Implement menu caching utility to optimize API calls	2
2026-01-09	JakeConal	feat: Implement password reset functionality with token validation	10
2026-01-09	JakeConal	feat: track individual item status within an order	2
2026-01-09	JakeConal	feat: Email verification link sent on signup in customer app	15

Date	Author	Commit Message	Files Changed
2026-01-09	JakeConal	feat: Add password complexity validation and strength indicator for login and profile pages	4
2026-01-09	JakeConal	feat: Implement password complexity validation for change password and sign-up processes	4
2026-01-09	JakeConal	feat: Integrate order history API and enhance order data structure in profile page	2
2026-01-09	JakeConal	feat: Add FAQ section with collapsible answers in profile page	1
2026-01-09	JakeConal	feat: Enhance profile picture handling with improved error handling and caching	4
2026-01-09	JakeConal	feat: Implement change password	9
2026-01-09	JakeConal	feat: Implement order tracking and payment functionality	15
2026-01-08	JakeConal	fix: correct query builder logic for fetching reviews by restaurant ID and reset review state on item ID change	2
2026-01-08	JakeConal	refactor: remove favorites page and update menu item fetching logic	4
2026-01-08	JakeConal	feat: enhance cart and menu pages with improved UI, URL state management, and item details display	2
2026-01-08	JakeConal	feat: enhance photo handling in menu service and item detail page; add swipe functionality for photo gallery	3
2026-01-08	JakeConal	feat: implement review module with CRUD operations and integrate with frontend	9
2026-01-03	Toan Huynh	chore: update .gitignore to include .idea and frontend/.next directories	1
2026-01-07	JakeConal	feat: Add sort dropdown, improve loading/error states, and refine item display styles	2
2026-01-07	JakeConal	feat: Revamp Menu and Bottom Navigation UI; enhance user experience with modern design elements	15
2026-01-07	JakeConal	refactor: layout and login page styles; update menu item handling	12
2026-01-02	JakeConal	feat: Enhance UI and UX for Menu and Bottom Navigation	13

Date	Author	Commit Message	Files Changed
2026-01-02	JakeConal	feat: update MenuItem to use primaryPhotoUrl and adjust related types	8
2026-01-01	JakeConal	refactor: remove unused UI components and API services	7
2026-01-01	JakeConal	fix: remove redundant Query in qr.service.ts and table.service.ts	4
2026-01-01	JakeConal	feat: Add signup and support pages with authentication and menu context	11
2025-12-31	JakeConal	feat: Add Customer entity and integrate with authentication service	8
2025-12-31	JakeConal	feat: Implement customer authentication and profile management	2
2025-12-31	JakeConal	feat: implement photo upload and retrieval for menu items	14
2025-12-31	JakeConal	feat: implement google login in Admin app	11
2025-12-28	JakeConal	fix: menu-item api just return 1 item only	2
2025-12-28	JakeConal	feat: initial customer app	15
2025-12-28	JakeConal	refactor: separate admin app and customer app	12
2025-12-27	Nam Anh	perf(table): use local QR generator	5
2025-12-27	HieuNguyen	Delete backend/.env.local	1
2025-12-27	JakeConal	feat: add authentication context and login/signup functionality	18
2025-12-27	JakeConal	refactor: integrate Admin and Customer app into 1 project	10
2025-12-27	Nam Anh	fix(toast): use `react-hot-toast`	4
2025-12-27	Nam Anh	refactor(api): modularize routes	22

Date	Author	Commit Message	Files Changed
2025-12-27	Nam Anh	feat(menu): add route	15
2025-12-26	Toan Huynh	feat: update controller paths to include API versioning and add findOne methods for MenuItem and ModifierGroup	8
2025-12-25	JakeConal	feat: add popularity score to MenuItem and implement incrementPopularity method in MenuItemService	3
2025-12-25	JakeConal	feat: implement MenuService and update MenuController to handle menu retrieval with query parameters	5
2025-12-25	JakeConal	fix: correct endpoint path for ModifierGroupController from '/modifier-group' to '/modifier-groups'	2
2025-12-25	JakeConal	feat: add ModifierGroup and ModifierOption modules with controllers, services, DTOs, and schemas	14
2025-12-25	JakeConal	feat: add MenuItemPhoto module with controller, service, schema, and tests	12
2025-12-25	JakeConal	feat: add MenuItem module with controller, service, DTOs, and schema	10
2025-12-25	JakeConal	feat: implement authentication module with JWT and user management	18
2025-12-25	JakeConal	fix: remove unused MenuCategory service and controller from AppModule	2
2025-12-25	JakeConal	feat: implement MenuCategory module with controller, service, and DTOs	11
2025-12-20	Toan Huynh	fix: exclude test files from production build	1
2025-12-20	Toan Huynh	fix: move @nestjs/cli and typescript to dependencies for Heroku build	1
2025-12-20	Toan Huynh	feat: update Procfile to include build step and add scripts section in package.json	2
2025-12-20	Toan Huynh	Add Procfile for Heroku deployment	1
2025-12-20	JakeConal	feat: update handleCreate function to accept both CreateTableDto and UpdateTableDto	2
2025-12-20	JakeConal	feat: update CORS configuration and improve code formatting in main.ts and page.tsx	2

Date	Author	Commit Message	Files Changed
2025-12-20	HieuNguyen	Merge pull request #1 from JakeConal/feature/table	12
2025-12-20	HieuNguyen	Merge branch 'main' into feature/table	10
2025-12-20	Toan Huynh	feat: add frontend and backend environment files to .gitignore	1
2025-12-20	Toan Huynh	feat: implement table page	15
2025-12-20	Nam Anh	fix: use yarn typescript	2
2025-12-20	JakeConal	fix: table in database not found	1
2025-12-19	Nam Anh	refactor: use proper import syntax	4
2025-12-19	Nam Anh	feat: switch to yarn & implement QR code	22
2025-12-19	JakeConal	feat(table): add TypeORM integration for table schema	5
2025-12-19	JakeConal	feat(table): implement table management module with CRUD operations	12
2025-12-19	JakeConal	feat: add backend with NestJS	25
2025-12-18	JakeConal	feat: initialize frontend with Next.js, React, and Tailwind CSS; backend with NestJS	17
2025-12-16	JakeConal	feat: initialize smart restaurant project with Next.js, React, and Tailwind CSS	0